

NurseFlow — Enterprise System & Database Architecture Manual

Full Technical Reference | 12 Prisma Relational Database Models, Gemma 4 AI Simulator & Shift Break Engine

1. Executive Overview & Platform Objectives

NurseFlow is a full-stack healthcare scheduling and clinical simulation platform. Built using React 18, Node.js, Prisma ORM, SQLite, Socket.IO, and Google Gemma 4 AI, NurseFlow solves shift scheduling friction, nurse fatigue, and clinical communication challenges across hospital wards.

- **Duty Roster Optimization:** Standardized shift timings (Morning 7am-3pm, Evening 3pm-11pm, Night 11pm-7am) with non-overlapping card layouts.
- **Gemma 4 AI Patient Simulator:** Educational roleplay simulator with dynamic emotion tracking and 8-competency evaluations.
- **AI Break Lounge Games:** 100% non-medical relaxation games (Tick & Cross) for 10-minute shift breaks.
- **Real-Time Push Alerts:** Socket.IO WebSocket push notifications for emergency ward calls and swap approvals.

2. Full Database Schema Data Dictionary (All 12 Prisma Models)

NurseFlow utilizes **Prisma ORM** with a relational SQLite database (dev.db). Below is the comprehensive technical breakdown of all 12 database models, their fields, default values, and relational constraints.

1. User Model

Stores staff credentials, role permissions (NURSE, ADMIN, HEAD_NURSE), department assignments, and contact details.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
name	String	Full Name of Staff Member
email	String (Unique)	Login Credential Email
passwordHash	String	Bcrypt Hashed Password
role	String (Default: NURSE)	NURSE ADMIN HEAD_NURSE
employeeId	String (Unique)	Hospital Badge ID
departmentId	String (Optional)	Foreign Key to Department
phone	String (Optional)	Contact Mobile Number
avatar	String (Optional)	Profile Picture Path

2. Department Model

Represents hospital ward units (e.g., General Ward, Intensive Care Unit, Emergency, Pediatrics).

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
name	String (Unique)	Department Name
description	String (Optional)	Ward Description & Capacity

3. Shift Model

Defines hospital shift timings, color coding, and shift categories.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
name	String	Shift Name (Morning, Evening, Night)
startTime	String	Formatted Start Time (e.g., 7:00 AM)
endTime	String	Formatted End Time (e.g., 3:00 PM)
type	String	MORNING EVENING NIGHT OFF
color	String	Theme Color Code (green, yellow, purple, gray)

4. Roster Model

Stores daily duty assignments linking nurses, shifts, and departments.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
nurseld	String	Foreign Key to User (Cascade Delete)
shiftId	String	Foreign Key to Shift (Cascade Delete)
departmentId	String	Foreign Key to Department (Cascade Delete)
date	String	Duty Date (YYYY-MM-DD)
status	String (Default: SCHEDULED)	ON_DUTY SCHEDULED OFF COMPLETED
notes	String (Optional)	Handover Shift Instructions

5. LeaveRequest Model

Tracks leave applications submitted by nurses and review decisions by administrators.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
nurseld	String	Foreign Key to User (Applicant)
leaveType	String	Annual Leave Sick Leave Emergency
fromDate	String	Start Date (YYYY-MM-DD)
toDate	String	End Date (YYYY-MM-DD)
reason	String	Application Reason Text
status	String (Default: PENDING)	PENDING APPROVED REJECTED

reviewedBy	String (Optional)	Foreign Key to Admin User
reviewedAt	DateTime (Optional)	Timestamp of Review Decision

6. ShiftSwapRequest Model

Manages nurse-to-nurse shift exchange applications.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
requesterId	String	Foreign Key to User (Requesting Nurse)
targetNurseId	String	Foreign Key to User (Peer Nurse)
originalShiftId	String	Foreign Key to Original Shift
requestedDate	String	Target Shift Date (YYYY-MM-DD)
reason	String	Swap Justification Note
status	String (Default: PENDING)	PENDING APPROVED REJECTED
reviewedBy	String (Optional)	Foreign Key to Admin Reviewer

7. Notification Model

Stores user-specific push notifications and emergency ward alerts.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
userId	String	Foreign Key to User
title	String	Notification Header Title
message	String	Push Notification Content
type	String (Default: INFO)	SHIFTS LEAVE SWAP ROSTER ALERT INFO
isRead	Boolean (Default: false)	Read/Unread State Flag

8. CommunicationScenario Model

Defines AI simulation roleplay scenarios and character personalities.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
title	String	Scenario Header Title
category	String	Patient Family Elderly Emergency
description	String	Detailed Context & Medical Background
characterRole	String	Patient Family Member Elderly Patient
personality	String	Anxious Hostile Confused Demanding
difficulty	String (Default: BEGINNER)	BEGINNER INTERMEDIATE ADVANCED
objectives	String	JSON Array String of Learning Goals

9. CommunicationSession Model

Tracks active and completed AI roleplay sessions.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
userId	String	Foreign Key to User (Nurse)
scenarioId	String	Foreign Key to CommunicationScenario
characterRole	String	Roleplayed Character Persona
difficulty	String	Simulation Complexity Level
status	String (Default: ACTIVE)	ACTIVE COMPLETED ABANDONED
overallScore	Int (Optional)	Final Overall Score (0-100)

10. CommunicationMessage Model

Stores individual dialogue turns and emotion states in roleplay chats.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
sessionId	String	Foreign Key to CommunicationSession
role	String	NURSE or PATIENT
content	String	Transcribed Dialogue Text
emotion	String (Default: Calm)	Calm Worried Confused Frustrated Angry Reassured

11. CommunicationAnalysis Model

Stores 8-competency scores and AI clinical feedback reports.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
sessionId	String (Unique)	Foreign Key to CommunicationSession
overallScore	Int	Calculated Overall Score (0-100)
empathyScore	Int	Empathy & Compassion Score
activeListeningScore	Int	Active Listening Score
clarityScore	Int	Communication Clarity Score
professionalismScore	Int	Professional Composure Score
emotionalIntelligenceScore	Int	EQ & Self-Awareness Score
deEscalationScore	Int	De-escalation Score
patientEngagementScore	Int	Patient Engagement Score
confidenceScore	Int	Clinical Confidence Score
strengths	String	JSON Array String of Key Strengths

improvementAreas	String	JSON Array String of Growth Areas
feedback	String	Comprehensive AI Coaching Summary Text

12. NurseGameScore Model

Tracks nurse wins, AI wins, draws, and earned wellness points in Tick & Cross game.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
userId	String	Foreign Key to User (Nurse)
gameType	String (Default: TIC_TAC_TOE)	Break Game Category
nurseWins	Int (Default: 0)	Total Nurse Victory Count
aiWins	Int (Default: 0)	Total Gemma 4 AI Victory Count
draws	Int (Default: 0)	Total Draw Count
pointsEarned	Int (Default: 0)	Total Shift Wellness Points Earned

3. Full REST API Endpoints Specification

Method	Endpoint Route	Functionality Description	Access
POST	/api/auth/login	Authenticate user credentials & return JWT token	Public
POST	/api/auth/register	Register new nurse or administrator account	Public
GET	/api/users	Fetch list of all staff nurses & employees	Admin
GET	/api/rosters	Fetch duty rosters (filter by date/nurse)	Auth
POST	/api/rosters	Assign shift roster entry to nurse	Admin
POST	/api/leave-requests	Submit leave application (Sick, Annual)	Nurse
POST	/api/shift-swaps	Request shift exchange with target nurse	Nurse
POST	/api/notifications/send-alert	Broadcast ward emergency push notification	Admin
GET	/api/communication-simulator/scenarios	Fetch roleplay scenarios list	Auth
POST	/api/communication-simulator/sessions	Initialize new AI roleplay session	Auth
POST	/api/communication-simulator/sessions/:id/messages	Send nurse dialogue & get AI roleplay reply	Auth
POST	/api/communication-simulator/sessions/:id/end	End session & generate 8-competency evaluation	Auth
POST	/api/games/tic-tac-toe/move	Compute Gemma 4 AI move (Minimax)	Auth
POST	/api/games/tic-tac-toe/score	Save match outcome & update wellness points	Auth